

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester IV)

MERN Stack

Practical – I

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1.Environment Setup and JavaScript Basics

a. Install Node.js and Visual Studio Code, and verify installation.

```
PS C:\Users\Sagar Kandari\OneDrive\Documents\MERN> node -v
v24.17.0
PS C:\Users\Sagar Kandari\OneDrive\Documents\MERN> npm -v
```

b. Create a simple Node.js file and run it using the terminal.

CODE:-

```
JS prac1b.js
1 console.log("welcome");
2 console.log("sagar");
```

OUTPUT:-

```
PS C:\Users\Sagar Kandari\OneDrive\Documents\MERN> node prac1b.js
welcome
sagar
```

c. Write a Node.js script demonstrating basic JavaScript commands (console log, variables, and arithmetic operations).

CODE:-

```
JS prac1c.js > ...
1  let num1 = 10;
2  let num2 = 20;
3
4  console.log("First number = ", num1);
5  console.log("Second number = ", num2);
6
7  console.log("Addition = ", num1 + num2);
8  console.log("Subtraction = ", num1 - num2);
9  console.log("Multiplication = ", num1 * num2);
10 console.log("Division = ", num1 / num2);
11
12 console.log("Sagar Kandari T010");
```

OUTPUT:-

```
Node.js v24.17.0
PS C:\Users\Sagar Kandari\OneDrive\Documents\MERN> node prac1c.js
First number = 10
Second number = 20
Addition = 30
Subtraction = -10
Multiplication = 200
Division = 0.5
Sagar Kandari T010
```

d. Create and run a simple program using variables and functions.

CODE:-

```
JS prac1d.js > ...
1  let num = 15;
2  let num2 = 10;
3
4  function Add(a, b) {
5      return a + b;
6  }
7
8  function Sub(a, b) {
9      return a - b;
10 }
11
12 console.log("Addition = " + Add(num, num2));
13 console.log("Subtraction = " + Sub(num, num2));
14 console.log("Sagar Kandari T010");
```

OUTPUT:-

```
PS C:\Users\Sagar Kandari\OneDrive\Documents\MERN> node prac1d.js
Addition = 25
Subtraction = 5
Sagar Kandari T010
```

e. Demonstrate use of operators and control statements in JavaScript

CODE:-

```
JS prac1e.js > ...
9
10 let day = 2;
11 switch (day) {
12     case 1:
13         console.log("Monday");
14         break;
15     case 2:
16         console.log("Tuesday"); // fixed spelling
17         break;
18     case 3:
19         console.log("Wednesday");
20         break;
21     default:
22         console.log("Invalid day");
23 }
24
25 console.log("Sagar Kandari T010");
26
```

OUTPUT:-

```
PS C:\Users\Sagar Kandari\OneDrive\Documents\MERN> node prac1e.js
Number 1 is greater 15
Tuesday
Sagar Kandari T010
```

f. Write programs using loops and template literals.

CODE:-

```
JS prac1f.js > ...
1  let num = 5;
2  console.log(`Multiplication Table of ${num}`);
3
4  for (let i = 1; i <= 10; i++) {
5      console.log(`${num} x ${i} = ${num * i}`);
6  }
7
8  console.log("Sagar Kandari T010");
9
```

OUTPUT:-

```
PS C:\Users\Sagar Kandari\OneDrive\Documents\MERN> node prac1f.js
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
Sagar Kandari T010
```